

# Demo runbook

## Visitor Counting Console

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### Demo runbook

Running the visitor console for a demo, on any of the three machines it is set up on. Everything from §3 onwards is the same everywhere; only starting and stopping differ.

|                | machine                                                                                      | how it runs                                     |
|----------------|----------------------------------------------------------------------------------------------|-------------------------------------------------|
| <b>Ubuntu</b>  | the demo box with an NVIDIA card, repo in e.g. <code>~/Documents/cv-demo-humanmonitor</code> | a systemd service, <code>visitor-console</code> |
| <b>Windows</b> | a demo PC with an NVIDIA card, repo in e.g. <code>C:\cv-demo-humanmonitor</code>             | <code>run_web.bat</code> in a terminal window   |
| <b>macOS</b>   | Apple Silicon laptop, the walkthrough machine                                                | <code>./run_web.sh</code> in Terminal           |

The console is on **port 8780** on all three.

Two ways to feed it, on all three:

|                                     | when                          |
|-------------------------------------|-------------------------------|
| <b>live</b> — the RTSP cameras      | showing the real installation |
| <b>video</b> — the five saved clips | showing the features          |

**Demonstrate on the clips, not on live cameras.** A camera pointed at a quiet corridor produces an empty frame, and an empty frame demonstrates nothing: the counters sit still and there is no way to show that any of it works. The clips are busy on purpose, so the numbers move while you are talking.

**macOS is the weakest of the three.** No CUDA, so the pipeline runs on the Apple GPU: three 720p files give roughly 5-10 pipeline fps. Fine for a walkthrough on a laptop, not for standing in front of a customer with live cameras. Use the Ubuntu box for that.

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## 1. Fifteen minutes before

The same three checks everywhere — only the path to the interpreter differs:

|                | interpreter                           |
|----------------|---------------------------------------|
| Ubuntu / macOS | <code>.venv/bin/python</code>         |
| Windows        | <code>.venv\Scripts\python.exe</code> |

### Ubuntu / macOS

```
cd ~/Documents/cv-demo-humanmonitor      # macOS: wherever the clone is
git fetch origin
git status -sb                            # want: "## main...origin/main" and nothing else
ls -lh sample_data/videos/                # five clips, ~110 MB total
```

### Windows

```
cd /d C:\cv-demo-humanmonitor
git fetch origin
git status -sb
dir sample_data\videos
```

`git status -sb` answers both questions at once. [behind N] on the first line means this machine is missing N commits — `git pull`. Any lines beneath it are local changes; if they are not yours, see the reset below. (This used to name a specific commit to check against, which went stale the moment anything was pushed.)

If the clips are missing, or are ~130 bytes:

```
.venv/bin/python download_demo_videos.py    # Ubuntu / macOS
.venv\Scripts\python.exe download_demo_videos.py  :: Windows
```

They are git-ignored on purpose, so downloading them never dirties the tree.

If `git status` is not empty and the changes are not yours — most likely an interrupted `git pull` — put the tree back in step:

```
git fetch origin && git reset --hard origin/main
```

That only touches tracked source. Your cameras, trigger lines, counting areas, hours, history and recordings all live in git-ignored files and are untouched.

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## 2. Starting it

### Ubuntu — the demo box

Live cameras run as the service:

```
sudo systemctl restart visitor-console
systemctl is-active visitor-console      # expect: active
```

For the saved clips, stop the service first — both want port 8780:

```
sudo systemctl stop visitor-console
./run_web.sh --mode video --capacity 150
```

Leave that terminal open; `Ctrl+C` stops it. Afterwards: `sudo systemctl start visitor-console`. To run the clips *alongside* the live service, give them a second port and open `http://localhost:8781`:

```
./run_web.sh --mode video --port 8781
```

## Windows

No service — it runs in a terminal window that stays open:

```
run_web.bat --mode video --capacity 150
run_web.bat --mode live --capacity 150
```

`Ctrl+C` in that window stops it; closing the window also does. On the first start Windows Defender Firewall asks whether to allow Python on private networks — **Allow** if another PC on the LAN should open the console.

If the PC is set up to start the console at logon (Task Scheduler task `VisitorConsole`, §4.5 of `DEPLOYMENT.md`), that instance already holds 8780; stop it first or use `--port 8781`:

```
Stop-ScheduledTask -TaskName VisitorConsole
```

## macOS

```
./run_web.sh --mode video --capacity 150
./run_web.sh --mode live --capacity 150
```

`Ctrl+C` stops it. If `pipeline fps` is low, `--device cpu` is sometimes faster than the Apple GPU for the classifier, and `--no-demographics` turns the male/female and adult/child work off entirely — worth knowing, but it also turns off half of what the demo is about, so prefer the clips at a smaller `--stream-width 640`.

## Then, in the browser, on all three

Open <http://localhost:8780> and press **Ctrl+Shift+R** (**Cmd+Shift+R** on macOS). The panel changes often and the browser will otherwise serve a cached stylesheet — if the layout does not match this runbook, the refresh did not take.

The **building day set-up** screen appears first: confirm the date, opening and closing times, capacity and starting occupancy, then **Start monitoring**. It asks once per day.

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## 3. What to show, in order

1. **The panel in one view** — VISITORS TODAY, CURRENT OCCUPANCY, CAMERAS DEPLOYED side by side. Every camera gets its own row in the third column; past five the list scrolls on its own rather than pushing anything else off the panel.
2. **The two figures are different things** — visitors today counts everyone who entered; current occupancy is who is inside now. Each box sums to the two tiles beneath it.
3. **The icons** — male/female and adult/child are told apart by shape, so they still read in greyscale or from across a room.
4. **Per camera** — click a card under CAMERAS DEPLOYED and the whole panel follows that entrance; click it again for the whole site.
5. **On the frames** — the same numbers are drawn into each camera image, so a recorded clip carries them.

6. **Swap a feed** — hover a camera panel and press **v**; **Shift+V** swaps all of them. Also the **> live / > video** button on each panel.
7. **Trigger lines** — *calibrate*, drag an end point, *save*. Two things on the overlay are worth naming before anyone asks: the faint dashed box is where the *picture* ends (the panel is wider than the frame, so the image is letterboxed inside it), and the faded dashed extension past each end of the line is the **8 % of its length that still counts** — drawn so the reach is not a surprise in the numbers. Placing a line within 4 % of the frame edge warns: people can cross it before the tracker has established them.
8. **Drawn areas** — *calibrate*, then *draw area*, pick what the area is for from the dropdown next to it, then **drag a box** across the picture (or click two opposite corners). Drag when the box needs to reach an edge: the button strip sits on top of the picture, and only the *start* of a drag has to miss it. The box is dashed on the frame in the colour of its role:

| role                               | what it does                                                        | line may sit outside?         |
|------------------------------------|---------------------------------------------------------------------|-------------------------------|
| <b>count only inside</b> (orange)  | only people inside are tracked at all — outside the box is dimmed   | <b>no</b> — a warning appears |
| <b>ignore inside</b> (red)         | people inside are skipped; use it on a window, a mirror or a poster | yes                           |
| <b>classify only inside</b> (blue) | everyone is counted, but only those inside are sexed and aged       | yes                           |

*classify only inside* is the one to demonstrate for accuracy: put it where people are large and well lit, and the crop budget stops being spent on distant figures. Tracks outside it still count, they just read **??**. *clear area* puts the whole frame back. A box has to be at least **5 % of the frame** on each side; smaller than that is refused, since it is more likely a stray click than an area anyone meant to draw.

While a box is being drawn the buttons that do not belong to that flow — *flip IN*, *clear*, *delete line*, the feed swap and *record* — grey out on purpose. They sit on the picture, and a corner placed near the bottom of the frame used to hit one of them instead. Areas save to `config/lines.json` beside the trigger lines, so anything set up in rehearsal is still there after a restart.

9. **What the area holds right now** — a counting area reports its own population without waiting for anyone to cross the line. The box carries it on the frame (**COUNTING AREA / 3 HERE 1M 2F 3A 0C**) and the panel repeats it as **in area now**. A trailing **?** is people the classifier has not decided yet, so the splits always add up to the total. This is a *presence* figure, not a crossing figure: it goes up and down as people walk through shot, and it is the one to point at when the demo starts mid-scene and the crossing counters are still at zero. With no area drawn the same line reads **in view now** and covers the whole frame.
10. **Capacity alarm** — *site*, set capacity below the current count. The panel banner, the alert row, the camera ribbon and that camera's card all turn red.
11. **Area capacity** — [settings] on a camera, set an area capacity of 3. That camera alarms on its own. Note this is a crossing figure and is not the same as *in area now* above.
12. **Record an event** — *\* record* saves 10 s before and 20 s after with a note, into `output/events/` and the history.

13. **History** — [expand] expands the console: alerts, cameras, crossings, hourly and closing totals, all written to `output/history.csv`.

#### 4. If something goes wrong

| symptom                                                                         | cause and fix                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| address already in use                                                          | something already holds 8780. <b>Ubuntu:</b> the service — <code>systemctl is-active visitor-console</code> , then use it or <code>sudo systemctl stop visitor-console</code> . <b>Windows:</b> the logon task — <code>Stop-ScheduledTask -TaskName VisitorConsole</code> , or <code>netstat -ano   findstr :8780</code> for the PID. <b>macOS:</b> <code>lsof -ti:8780</code> . Or just add <code>--port 8781</code> |
| layout looks like an older version                                              | cached CSS — <b>Ctrl+Shift+R</b> , or <b>Cmd+Shift+R</b> on macOS                                                                                                                                                                                                                                                                                                                                                     |
| a camera shows <b>OFFLINE</b>                                                   | the stream is unreachable. The card hides its counts on purpose: the numbers it had were frozen at the moment the link dropped                                                                                                                                                                                                                                                                                        |
| connect hangs when adding a camera                                              | should not happen since 4417398 — opens are capped at 5 s. If it does, the box is on an older commit                                                                                                                                                                                                                                                                                                                  |
| everyone on one camera reads ??                                                 | a <i>classify only inside</i> area that nobody walks through, or one covering only distant figures — move it, enlarge it, or <i>clear area</i>                                                                                                                                                                                                                                                                        |
| a camera counts nothing after an area was drawn                                 | the area is <i>count only inside</i> and the trigger line is not inside it — people are first seen already at the line. Either enlarge the area to cover the whole line with room on both sides, or switch the role to <i>classify only inside</i> . Saving it warns about this                                                                                                                                       |
| counts negative, or 0 <b>inside</b> with a note                                 | the clips start mid-scene, so people leave through lines they were never seen entering. <b>reset counters</b> in the footer, or set a starting occupancy on the set-up screen                                                                                                                                                                                                                                         |
| v does nothing, toast says no paired feed                                       | that slot has no counterpart. <code>config/sources.json</code> needs <code>live</code> and <code>video</code> lists of the same length — slot <i>i</i> pairs with the other list's entry <i>i</i>                                                                                                                                                                                                                     |
| a camera added on the page will not connect, and the password has symbols in it | @ : \$ % & and spaces are encoded for you - paste them as they are. / ? # must be typed as %2F %3F %23. If you passed the URL on a command line instead, single-quote it: \$\$ unquoted is the shell's process id                                                                                                                                                                                                     |
| cameras read <b>Camera 1 / 2 / 3</b>                                            | no display names set. [settings] on a panel, type a name, connect. It persists                                                                                                                                                                                                                                                                                                                                        |
| git pull refused                                                                | see §1                                                                                                                                                                                                                                                                                                                                                                                                                |

| symptom                                               | cause and fix                                                                                                                                                                     |
|-------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| pipeline fps in low single figures                    | expected on macOS (no CUDA) — see the note at the top. On Ubuntu or Windows it means the GPU is not being used: check the log's first lines for <code>CUDA available: True</code> |
| Windows: SmartScreen or antivirus blocks the launcher | allow <code>.venv\Scripts\python.exe</code> ; keep the folder out of a OneDrive-synced path                                                                                       |
| macOS: OpenCV windows never appear                    | <code>run_demo.sh</code> needs a logged-in desktop session, not SSH. The browser console is the headless option                                                                   |

**Logs.** Ubuntu service: `journalctl -u visitor-console -f`. Everywhere else: the terminal window it is running in.

**Health check**, the same on all three:

```
curl -s http://localhost:8780/api/state | head -c 300
```

```
curl.exe -s http://localhost:8780/api/state           # Windows PowerShell
```

Look for "error": null, each camera "status": "LIVE" or "FILE", and a pipeline\_fps above zero.

## 5. Afterwards

### Ubuntu

```
sudo systemctl start visitor-console           # if you stopped it for the clips
```

**Windows** — close the terminal window, or `Start-ScheduledTask -TaskName VisitorConsole` if the PC normally runs it at logon.

**macOS** — `Ctrl+C` in Terminal.

On all three: the day's numbers are in `output/history.csv`; recorded clips and snapshots in `output/events/`. Neither is in git — copy them off if they are needed. Trigger lines and counting areas stay in `config/lines.json`, so the next demo starts calibrated.

Counters reset at the opening time set on the set-up screen.